

1. Write a program to traverse a 2D matrix in a spiral order.

Input:

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1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16
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Output: 1 2 3 4 8 12 16 15 14 13 9 5 6 7 11 10

2. Write a program that will take one string as input. The program will then remove vowels a, e, i, o, and u(in lower or upper case) from the string. If there are two or more vowels that occur together then the program shall ignore all of those vowels.

Example 1

Input: Cat

Output: Ct

Example 2

Input: Computer

Output: Cmpuutr

3. Write number in expanded form.

Input 1: 12

Output 1: "10 + 2"

Input 2: 70304

Output 2: "70000 + 300 + 4"

4. Find the extra element and its index

Input 1: [10, 20, 30, 12, 5],[10, 5, 30, 20]

Output 1: 12 is the extra element in array 1 at index 3

Input 2: [-1, 0, 3, 2],[3, 4, 0, -1, 2]

Output 2: 4 is the extra element in array 2 at index 1

5. Given an integer array nums, return an array output such that output[i] is equal to the product of all the elements of nums except nums[i].

Input 1: [1,2,3,4]

Output 1: [24,12,8,6]

Input 2: [-1,1,0,-3,3]

Output 2: [0,0,9,0,0]